

React

```
import Usememoo from "./Usememoo";
import ConsumeContext from "./context/ConsumeContext";
import { Uerprovider } from "./context/Usercontext";
import CustomHook from "./CustomHook";
import ControlledComponent from "./ControlledComponent";
import AxiosComponent from "./axios/AxiosComponent";
import Conditional from "./Conditional";
import { BrowserRouter, Link, Route, Routes } from "react-router-dom";
import Home from "./pages/Home";
import About from "./pages/About";
import Header from "./pages/Header";
import Products from "./pages/Products";
import ProductPage from "./pages/ProductPage";

function App() {
  return (
    <>
      <h2>Header</h2>
      <Header />

      { /* <Usememoo/> */ }
      { /* <ConsumeContext/> */ }
      { /* <CustomHook/> */ }
      { /* <ControlledComponent/> */ }
      { /* <AxiosComponent/> */ }
      { /* <Conditional/> */ }

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
        <Route path="/product" element={<Products />} />
        <Route path="/product/:id" element={<ProductPage />} />
      </Routes>

      <h2>Footer</h2>
    </>
  );
}

export default App;
```

```
import axios from "axios";
import React, { useEffect, useState } from "react";
import { Link, Links } from "react-router-dom";
```

```

const Products = () => {
  const [data, setData] = useState([]);

  const fetchData = async () => {
    const result = await axios.get("https://dummyjson.com/products");
    console.log(result.data.products);
    setData(result.data.products);
  };
  useEffect(() => {
    fetchData();
  }, []);
  return (
    <div
      style={{
        display: "flex",
        justifyContent: "center",
        alignItems: "center",
        flexWrap: "wrap",
        gap: "20px",
      }}
    >
      {data.map((ele) => {
        return (
          <Link to={` /product/${ele.id}`} >
            <div
              key={ele.id}
              style={{ border: "2px solid black", padding: "20px" }}
            >
              <p>{ele.title}</p>
              <img
                src={ele.images[0]}
                alt=""
                style={{ width: "100px", height: "100px" }}
              />
            </div>
          </Link>
        );
      })}
    </div>
  );
};

export default Products;

```

```

import axios from "axios";

```

```

import React, { useEffect, useState } from "react";
import { useParams } from "react-router-dom";

const ProductPage = () => {
  const { id } = useParams();
  console.log(id);

  const [data, setData] = useState({});
  const fetchData = async () => {
    const result = await axios.get(`https://dummyjson.com/products/${id}`);

    console.log(result.data);
    setData(result.data);
  };

  useEffect(() => {
    fetchData();
  }, []);

  return (
    <div>
      <h2>Product page</h2>
      <div>
        <h1>{data.title}</h1>
        <img
          src={data.images}
          style={{ width: "100px", height: "100px" }}
          alt=""
        />
        <p>{data.description}</p>
      </div>
    </div>
  );
};

export default ProductPage;

```

```

import React from "react";

const Home = () => {
  return (
    <div>
      <h1>Home</h1>
    </div>
  );
};

```

```
export default Home;
```

```
import React from "react";
import { Link } from "react-router-dom";

const Header = () => {
  return (
    <div>
      <span>
        {" "}
        <Link to={"/"}>Home</Link>{" "}
      </span>
      <span>
        <Link to={"/about"}>About</Link>
      </span>

      <span>
        <Link to={"/product"}>Product</Link>
      </span>
    </div>
  );
};

export default Header;
```

```
import React from 'react'

const About = () => {
  return (
    <div>
      <h1>About</h1>
    </div>
  )
}

export default About
```

CONTEXT

```
import React, { useContext } from "react";
import Usercontext from "../Usercontext";

const ConsumeContext = () => {

  const { name, age } = useContext(Usercontext);
  return (
    <div>
      <h2>
        {name} - {age}
      </h2>
    </div>
  );
};

export default ConsumeContext;
```

```
import { createContext } from "react";

const Usercontext = createContext();

export const Uerprovider = ({ children }) => {
  let obj = {
    name: "FIIT",
    age: 20,
  };
  return <Usercontext.Provider value={obj}>{children}</Usercontext.Provider>;
};

export default Usercontext;
```

REDUX

```
import { createSlice } from "@reduxjs/toolkit";

const userSlice = createSlice({
  name: "user",
  initialState: { name: "Guest", isLoggedIn: false },
  reducers: {
    login: (state, action) => {
      state.name = action.payload;
      state.isLoggedIn = true;
    },
    logout: (state) => {
```

```

        state.isLoggedIn = false;
    },
  },
});

export const {login, logout} = userSlice.actions;

export default userSlice.reducer;

```

```

import { configureStore } from "@reduxjs/toolkit";
import counterSlice from "./counterSlice";
import userSlice from "./userSlice";

export const store = configureStore({
  reducer: {
    counter : counterSlice,
    user : userSlice
  }
});

```

```

import { createSlice } from "@reduxjs/toolkit";

const counterSlice = createSlice({
  name: "counter",
  initialState: { data: 0 },
  reducers: {
    increment: (state) => {
      state.data += 1;
    },
    decrement: (state) => {
      state.data -= 1;
    },
    reset: (state) => {
      state.data = 0;
    },
  },
});

export const { increment, decrement, reset } = counterSlice.actions;
export default counterSlice.reducer;

```

```

import { useState } from "react";
import reactLogo from "../assets/react.svg";
import viteLogo from "../assets/vite.svg";
import heroImg from "../assets/hero.png";
import "../App.css";
import { useDispatch, useSelector } from "react-redux";
import { decrement, increment, reset } from "../store/counterSlice";
import { login } from "../store/userSlice";

function App() {
  const count = useSelector((state) => state.counter.data);
  const dispatch = useDispatch();

  const { name, isLoggedIn } = useSelector((state) => state.user);
  console.log(name, isLoggedIn);

  // zustand
  return (
    <>
      {count}
      <h2 className="text-3xl p-6 m-6 border-3 border-sky-300 text-yellow-600">Redux</h2>
      <h1>{name}</h1>

      <button onClick={() => dispatch(login("FIIT"))}>Login</button>

      <button
        onClick={() => {
          dispatch(increment());
        }}
      >
        Increment
      </button>

      <button onClick={() => dispatch(decrement())}>decrement</button>
      <button onClick={() => dispatch(reset())}>Reset</button>

      <button></button>
    </>
  );
}

export default App;

```

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'
import { Provider } from 'react-redux'
import { store } from './store/store.js'

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <Provider store={store}>

      <App />
    </Provider>
  </StrictMode>,
)
```

Theory

- **React Router**
 - BrowserRouter, Routes, Route → used for navigation between pages.
 - Link → navigates without reloading the page.
 - useParams → extracts dynamic route parameters (e.g., /product/:id).
- **Axios**
 - Used for API calls (get, post, put, delete).
 - Example: axios.get("https://dummyjson.com/products").
- **Context API**
 - createContext() → creates a context.
 - Provider → supplies data to child components.
 - useContext() → consumes context values.
- **Redux Toolkit**
 - createSlice() → defines state and reducers.
 - configureStore() → combines slices into a store.
 - useSelector() → reads state.
 - useDispatch() → dispatches actions.

Coding Round Questions

- React Router → Create a React app with routes /home, /about, and /products. Display different components for each route.
- Dynamic routing → Implement a route /product/:id that shows product details based on the ID.
- Axios fetch → Fetch product data from an API using Axios and display it in a list.
- Context API → Create a context that provides name and age to child components. Consume it using useContext().
- Redux counter → Implement a counter with Redux Toolkit that supports increment, decrement, and reset.
- Redux login → Create a Redux slice for user authentication with login and logout actions.

⚡ Machine Coding Round Challenge

Problem: E-Commerce App with React Router, Axios, Context, and Redux

Requirements:

1. Routing:

- / → Home page.
- /about → About page.
- /products → Product listing.
- /product/:id → Product details page.

2. Axios Integration:

- Fetch product list from API (<https://dummyjson.com/products>).
- Display product cards with title and image.
- On click, navigate to product details page.

3. Context API:

- Provide global user info (name, age) using Context.
- Consume it in a component to display user details.

4. Redux Toolkit:

- Implement a counter slice with increment, decrement, reset.
- Implement a user slice with login/logout.
- Display counter and user info in the app.

Expected Flow:

- User visits /products → sees product list.
- Clicks a product → navigates to /product/:id → sees product details.
- Context shows user info globally.
- Redux manages login state and counter.